

ELEC 221 Lecture 05

Impulse response stability and causality; CT Fourier series

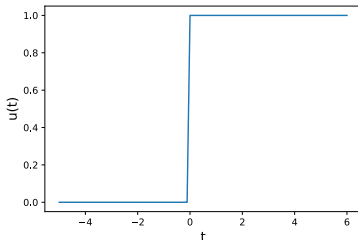
Thursday 24 September 2026

Announcements

- Tutorial Assignment 1 due tomorrow night at 23:59
- Quiz 3 Tuesday
- Tutorial 🧑 on Monday: practice problems

Last time

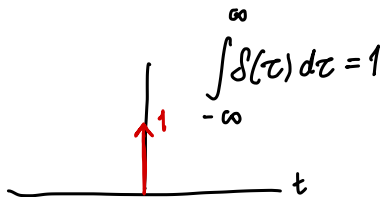
We defined the CT unit step, $u(t)$



$$u(t) = \begin{cases} 1 & t \geq 0 \\ 0 & t < 0 \end{cases}$$

And the CT unit impulse function, $\delta(t)$:

$$\delta(t) = \frac{du(t)}{dt}$$



Last time

DT conv sum: $y[n] = \sum_{k=-\infty}^{\infty} x[k] h[n-k]$

We introduced the CT convolution integral,

$$y(t) = \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau$$

If we know the impulse response of an LTI system, we can determine its response to any other signal.

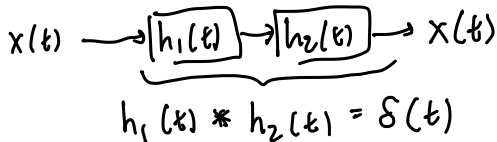
Last time

We saw how properties of the impulse response are related to the system's properties as a whole.

(Memory) For a system to be memoryless,

$$h(t) = K \delta(t) \quad h[n] = K \delta[n]$$

(Invertibility) If a system with impulse response $h(t)$ is invertible, there exists an *inverse system* with impulse response $h_i(t)$ s.t.



(Analogous for DT case)

Today

Learning outcomes:

- Use the impulse response to determine whether a system is stable or causal
- Use the convolution integral to show that complex exponentials are eigenfunctions of LTI systems
- Define and compute the system function (frequency response) of an LTI system
- Express a periodic CT signal as a Fourier series, and compute its Fourier coefficients

Impulse response and stability

Suppose we have a signal $x(t)$ with a bounded input, $|x(t)| \leq B$.
If the system is stable, the output should be bounded.

$$\begin{aligned}
 y(t) &= \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau \\
 &= \int_{-\infty}^{\infty} x(t-\tau) h(\tau) d\tau \\
 |y(t)| &= \left| \int_{-\infty}^{\infty} x(t-\tau) h(\tau) d\tau \right| \\
 &\leq \int_{-\infty}^{\infty} |x(t-\tau)| \cdot |h(\tau)| d\tau \\
 &\leq \int_{-\infty}^{\infty} B \cdot |h(\tau)| d\tau \\
 &= B \cdot \boxed{\int_{-\infty}^{\infty} |h(\tau)| d\tau}
 \end{aligned}$$

*must be
finite
to be stable.*

Impulse response and stability

As long as

$$\int_{-\infty}^{\infty} |h(\tau)| d\tau \quad \text{finite}$$

(i.e., $h(t)$ is *absolutely integrable*), the system is stable.

In DT, $h[n]$ must be *absolutely summable* to ensure stability:

$$\sum_{k=-\infty}^{\infty} |h[k]| \quad \text{finite}$$

Example/exercise: stability

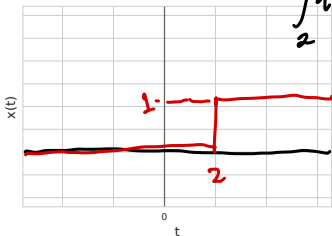
Consider systems A and B with impulse responses

$$h_A(t) = u(t-2)$$

not stable

Are they stable?

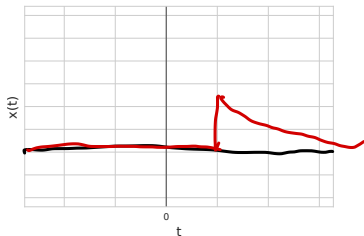
$$h_A(t) \quad \int_2^{\infty} 1 d\tau$$



$$h_B(t) = e^{-4t} u(t-2)$$

stable

$$h_B(t)$$



Example/exercise: stability

$$\begin{aligned}
 \int_{-\infty}^{\infty} |h(\tau)| d\tau &= \int_{-\infty}^{\infty} |e^{-4\tau} u(\tau-2)| d\tau \\
 &= \int_2^{\infty} |e^{-4\tau}| d\tau \\
 &= \int_2^{\infty} e^{-4\tau} d\tau \\
 &= -\frac{1}{4} e^{-4\tau} \Big|_2^{\infty} \\
 &= -\frac{1}{4} [0 - e^{-8}] \\
 &= \frac{1}{4} e^{-8}
 \end{aligned}$$

Impulse response and causality

Recall definition of causal signal and consider the convolution sum:

$$y[n] = \sum_{k=-\infty}^{\infty} x[n-k] h[k]$$

What properties does $h[n]$ need to have for system to be causal?

If k is neg: $x[n-k]$ in future

$$h[k] = 0 \quad k < 0$$

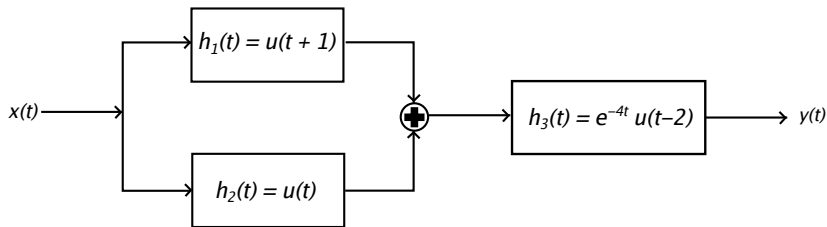
(Analogous holds for CT systems)

Final exercise

Homework

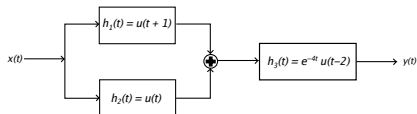
⇒ will take up in tutorial.

Consider the following combination of systems:

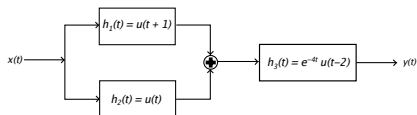


Is this system causal and/or stable?

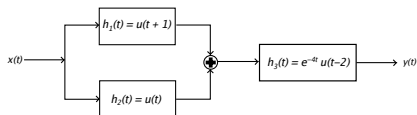
Final exercise



Final exercise



Final exercise



Review: complex exponential functions

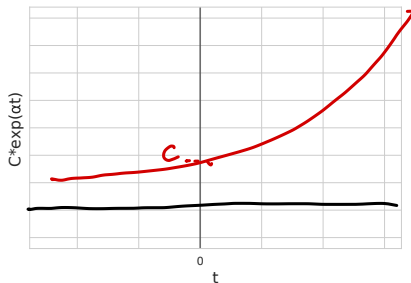
Most general form:

$$x(t) = C e^{\alpha t}$$

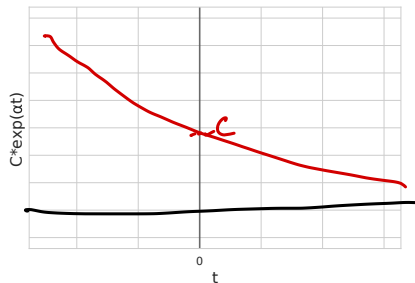
where α can be real or complex.

Case: both C and α are real-valued.

$\alpha > 0$



$\alpha < 0$



Review: complex exponential functions

Case: α is complex. Most generally, we can write

$$\alpha = a + bj \quad \alpha = |\alpha| e^{j\theta}$$

$$|\alpha| = \sqrt{a^2 + b^2}$$

Euler's relation allows us to write

$$e^{j\theta} = \cos \theta + j \sin \theta$$

$$\cos \theta = \frac{e^{j\theta} + e^{-j\theta}}{2}$$

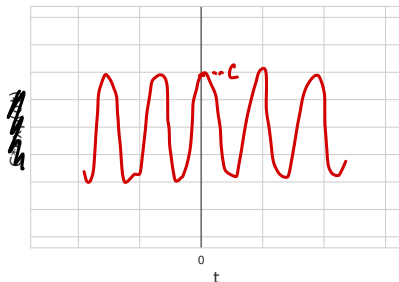
$$\sin \theta = \frac{e^{j\theta} - e^{-j\theta}}{2j}$$

Review: complex exponential functions

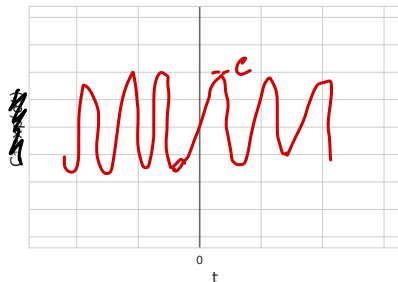
Case: $\alpha = j\omega$. Then,

$$x(t) = Ce^{\alpha t} = Ce^{j\omega t} \\ = C \cos \omega t + jC \sin \omega t$$

$\text{Re}[x(t)]$



$\text{Im}[x(t)]$

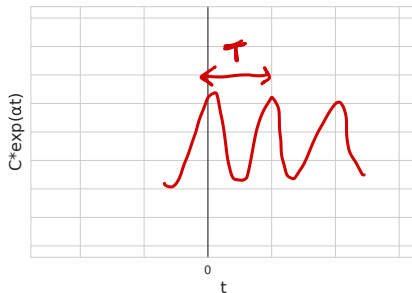


ω : frequency

Review: complex exponential functions

$x(t) = Ce^{j\omega t}$ is periodic:

$$x(t+T) = C e^{j\omega(t+T)} = \underbrace{C e^{j\omega t}}_{x(t)} \underbrace{e^{j\omega T}}_{=1} = x(t)$$



if freq is ω :

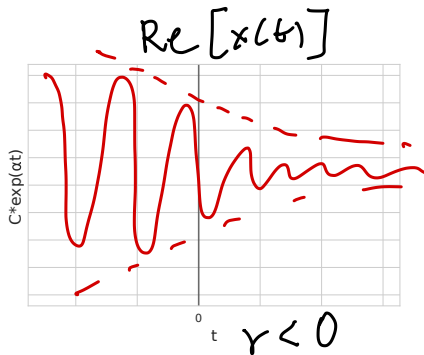
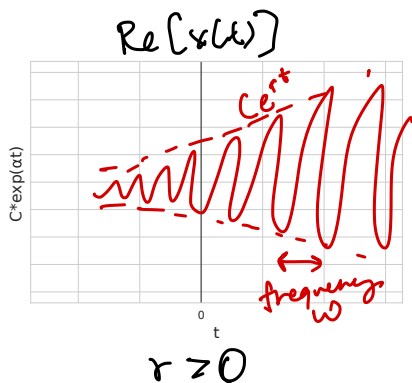
$$T = \frac{2\pi}{\omega}$$

$$\begin{aligned} e^{j\omega T} &= e^{j\omega \cdot \frac{2\pi}{\omega}} \\ &= e^{j \cdot 2\pi} \\ &= 1 \end{aligned}$$

Review: complex exponential functions

Case: $\alpha = r + j\omega$. Then,

$$s = r + j\omega. \text{ Then, } x(t) = C e^{\alpha t} = C e^{r t} e^{j\omega t}$$



LTI systems and complex exponential functions

Recall the convolution integral:

$$y(t) = \int_{-\infty}^{\infty} x(\tau) h(t - \tau) d\tau$$

What happens when $x(t)$ is a complex exponential signal?

LTI systems and complex exponential functions

Write $x(t) = e^{st}$. Then:

$$\begin{aligned} y(t) &= \int_{-\infty}^{\infty} x(t-\tau) h(\tau) d\tau \\ &= \int_{-\infty}^{\infty} e^{s(t-\tau)} h(\tau) d\tau \\ &= e^{st} \int_{-\infty}^{\infty} e^{-s\tau} h(\tau) d\tau \\ &= x(t) \cdot \int_{-\infty}^{\infty} e^{-s\tau} h(\tau) d\tau \end{aligned}$$

is e^{st} the only eigenfunction of LTI systems

eigenfunction!
h)

system function or transfer function

LTI systems and complex exponential functions

To summarize:

$$x(t) = e^{st} \rightarrow y(t) = H(s) e^{st}$$

Complex exponentials are **eigenfunctions** of LTI systems.

$H(s)$ is the **system function**.

$$H(s) = \int_{-\infty}^{\infty} e^{-s\tau} h(\tau) d\tau$$

...so what?

Recall that for LTI systems,

$$x(t) = \sum_i a_i x_i(t) \longrightarrow y(t) = \sum_i a_i y_i(t)$$

If all $x_i(t)$ are complex exponential signals, and

$$x(t) = e^{st} \longrightarrow y(t) = H(s) e^{st} = H(s) \cdot x(t)$$

then

$$x(t) = \sum_k c_k e^{s_k t} \longrightarrow y(t) = \sum_k c_k H(s_k) e^{s_k t}$$

The response is a superposition **of those same signals**, scaled by the system function.

System functions

Consider the limited set of signals¹:

$$x(t) = e^{st} = e^{j\omega t} \quad (s = j\omega)$$

$x(t)$ has frequency ω and period $T = 2\pi/\omega$.

When such signals are input into an LTI system, the system function, $H(s) = H(j\omega)$, is called the *frequency response*.

¹We will see the general case at the end of the course.

Exercise: frequency response

Suppose we have access to a system with impulse response

$$h(t) = \frac{1}{2}\delta(t-1) \quad H(s) = \int_{-\infty}^{\infty} e^{-s\tau} h(\tau) d\tau$$

1. What is the frequency response, $H(j\omega)$, of the system?

$$H(j\omega) = \int_{-\infty}^{\infty} e^{-j\omega\tau} \frac{1}{2}\delta(\tau-1) d\tau$$

$$= \frac{1}{2} e^{-j\omega}$$

$$x(t) = e^{j\omega t}$$

$$y(t) = \frac{1}{2} e^{-j\omega} \cdot e^{j\omega t}$$

$$= \frac{1}{2} e^{j\omega(t-1)}$$

Exercise: frequency response

Suppose we have access to a system with impulse response

$$H(j\omega) = \frac{1}{2}e^{-j\omega}$$

$$h(t) = \frac{1}{2}\delta(t-1)$$

$$y(t) = H(j\omega) \cdot x(t)$$

2. What is the output of the system given input

$$x(t) = e^{j\omega t}$$

$$x(t) = 2e^{2j\omega t} - 3e^{5j\omega t}$$

$$\begin{aligned} y(t) &= 2 \cdot H(2j\omega) \cdot e^{2j\omega t} - 3 H(5j\omega) e^{5j\omega t} \\ &= 2 \cdot \frac{1}{2} e^{-2j\omega} e^{2j\omega t} - 3 \cdot \frac{1}{2} e^{-5j\omega} e^{5j\omega t} \\ &= e^{2j\omega(t-1)} - \frac{3}{2} e^{5j\omega(t-1)} \end{aligned}$$

Exercise: frequency response

Suppose we have access to a system with impulse response

$$h(t) = \frac{1}{2}\delta(t - 1)$$

3. What is the output of the system given input

$$x(t) = \frac{2}{3}\cos(3\omega t)$$

*try it
yourself.*

The Fourier series

How can we express arbitrary signals using complex exponentials?

$$x(t) = \sum_k c_k e^{j\omega_k t} \rightarrow y(t) = \sum_k c_k H(j\omega_k) e^{j\omega_k t}$$

How to make signals look like this?!

The Fourier series

Consider

$$x(t) = e^{j\omega t}$$

It also has an infinite number of cousins called *harmonics*.

$$x_k(t) = e^{jk\omega t} = e^{jk \frac{2\pi}{T} t} \quad k = 0, \pm 1, \pm 2, \dots$$

ω is the **fundamental frequency**.

(Yes, these harmonics.)

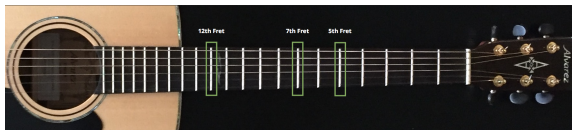
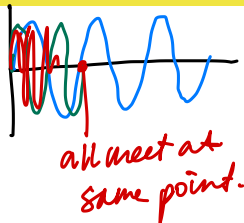


Image credit: <https://musiciantuts.com/guitar-harmonics/>

The Fourier series

We can create a superposition of all harmonics,

$$\sum_{k=-\infty}^{\infty} c_k e^{jk\omega t}$$



This signal is also periodic with period $T = 2\pi/\omega$.

Given $x(t)$ with period T ², we can write it as a **Fourier series**.



not T^2 , just footnote

$$x(t) = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega t}$$

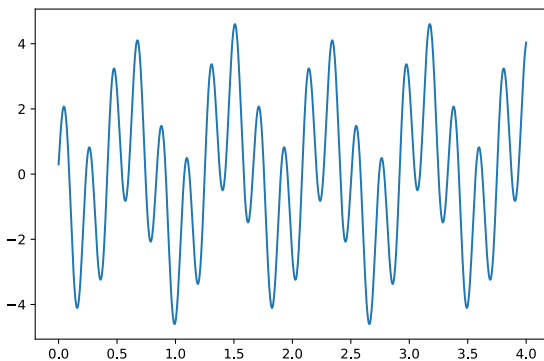
$\omega = \frac{2\pi}{T}$

The c_k (complex numbers) are its **Fourier coefficients**.

²There are some additional conditions; we will cover this next class

Going backwards

What if we are given a signal like this:



Can we determine the c_k ?

For next time

Content:

- Dirichlet conditions and the Gibbs phenomenon
- Properties of Fourier series and Fourier coefficients

Action items:

- ~~Assignment 1 is due tonight~~
- Tuesday quiz on this week's material

Recommended reading:

- From today's class: Oppenheim 1.3, 3.0-3.3
- Suggested problems: 1.6, 1.8, 1.10, 3.1, 3.3, 3.4, 3.13, 3.17, 3.22a,c
- For next class: Oppenheim 3.4-3.5